



Economies of Scale in the Real Estate Mutual Fund Industry

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Abstract

This paper investigates the role of scale in Real Estate Mutual Fund (REMF) performance. We test the impact of both fund-level and industry-level economies of scale on fund performance. We provide consistent evidence that industry size erodes REMF performance. After controlling for endogeneity concerns, we document an insignificant relation between fund size and performance. Taken together, these findings suggest that the rapid growth of the REMF industry over the past few decades has materially impacted active managers' ability to consistently outperform their passive benchmarks. As more capital flows into the industry, competition for alpha increases, and investment opportunities dwindle. This effect is stronger in funds who are especially active in seeking out those increasingly elusive opportunities. Specifically, the effect of industry size on alpha is particularly negative for funds with higher turnover ratios, expense ratios, and volatility of returns.

Keywords Real Estate Mutual Fund · Economies of Scale · REITs · Fund Performance · Alpha

Introduction

Perhaps the most central research question in the mutual fund literature is whether active managers can consistently outperform their passive benchmarks on a risk-adjusted basis. Although the case is not closed and the research area remains active,

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the data suggests the average manager in the diversified mutual fund space is unable to achieve alpha. For those that do outperform, there is limited evidence that outperformance persists. These findings are not limited to the academic community. The S&P Indices Versus Active (SPIVA) produces a bi-annual report tracking mutual funds versus active benchmarks. As of December 2019, they report that less than one in five large-cap mutual funds outpaced the S&P 500 over the past five years.¹

Our paper focuses on a subset of mutual funds that are specialized in the real estate sector, known as real estate mutual funds (REMFs). REMFs have experienced explosive growth in the past two decades as investors become increasingly aware of the benefits of a well-diversified portfolio. Access to commercial real estate as an alternative investment for a typical investor was historically difficult due to high capital requirements, high fees, lack of transparency, and illiquidity of the assets. Public listed real estate investment trusts (REIT) solved many of these issues and investor appetite for the securities followed, as evidenced by the growth of the industry from approximately \$28 billion in 1995 to \$1.3 trillion in 2019 (4,345%). REMFs, which invest predominantly in public listed REITs, have experienced even more dramatic growth over the same period. REMFs have grown from approximately \$1.5 billion in assets under management (AUM) in 1995 to \$111 billion in 2019 (7,543%).

REMFs are interesting to investigate because while the diversified mutual fund literature tends to document lackluster performance of active managers, findings related to REMF performance are less consistent. A seminal paper by Kallberg et al. (2000) finds that, contrary to the diversified mutual fund literature, REMF funds do generate alpha on average. The authors argue that due to the labor-intensive nature of visiting REIT properties and the uniqueness of the security, the cost of value-relevant information is higher, providing an advantage to active managers.² Several other papers echo this sentiment and document outperformance of REMFs (Gallo et al., 2000; Cici et al., 2011; Kaushik & Pennathur, 2012).

In contrast, other studies have documented a lack of outperformance of REMFs, particularly after fees (Chiang et al., 2008; Derwall et al., 2009; Hartzell et al., 2010; Chou & Hardin, 2014; MacGregor et al., 2020). As Ro and Gallimore (2014) note, REITs are well-known for being transparent compared to industrial firms. If the cost of information for REITs is particularly low due to their transparent nature, REMF managers may struggle to identify undervalued securities because their expertise may not provide a competitive advantage.

Hartzell et al. (2010) focus on the impact of the appropriate benchmark model to evaluate REMF performance and find no abnormal outperformance. However, even with their proposed benchmark model, they document positive alpha when testing their earlier sample period used in Kallberg et al. (2000). They offer the following

¹ <https://www.spindices.com/spiva/#/reports>.

² This sentiment is well-captured by a quote from David Lee, the portfolio manager of the T. Rowe Price Real Estate Funds following a period of outperformance: “While investors tend to look at REITs as a group, mutual fund managers know to slice the market into areas that behave differently during economic cycles.” Real Estate Funds Defy Forecasts with 27.6% 2010 Gain,” *USA Today*, January 6, 2011 (Kaushik & Pennathur, 2013).

potential explanation for these results: “The fact that there was abnormal performance in this earlier period but not in our later time period suggests that the increase in the number of mutual funds and other institutions investing in REITs may have diluted average fund performance.” (Hartzell et al., 2010, pg.125). Our paper, which focuses on how fund size and industry size impact REMF performance, provides a formal investigation of this assertion and potentially resolves the dispute in the literature.

While the diversified mutual fund literature has found that the industry is highly competitive and alpha is rarely consistently generated, the exact mechanism at work is less clear. Fund level and industry level economies of scale are theories put forth as potential mechanisms to explain this phenomenon. Although earlier papers in the diversified mutual fund literature produced competing hypotheses regarding fund level economies of scale, Berk and Green (2004) provide a seminal theoretical contribution to the topic.³ Their premise is that skilled funds with positive track records attract capital flows. As funds that outperform attract additional capital, their ability to capitalize on investment opportunities diminishes, explaining the lack of persistence in mutual fund performance.⁴

Pástor and Stambaugh (2012) are the first to develop a theory regarding industry-level diseconomies of scale. They argue that as more money chases limited investment opportunities, the ability to outperform diminishes. Importantly, similar to Berk and Green (2004) they seek to address the aforementioned puzzle of why an industry that does not seem to persistently beat the market continues to exist. It is not due to a lack of fund skill, but rather to the competitive allocation of investors capital to the mutual fund industry.

The fund-level and industry-level theories are not mutually exclusive. Although they are motivated somewhat similarly, they differ on a few key dimensions. Berk and Green (2004) argue that funds will shift from low performing funds to high performing funds, whereas Pástor and Stambaugh (2012) argue funds will shift in and out of the aggregate active management industry. If managers follow similar trading strategies, then it is more likely that the aggregate industry size would impact performance (Pástor et al., 2015). In contrast, if investment strategies are unrelated, only the capital flows at the fund level would erode that funds investment opportunity set.

REMFs have several unique features that provide reason to suspect the relation between fund and industry size on performance may be different from that in the diversified mutual funds literature. First, given a significantly smaller investable universe largely constrained to a single industry, it is plausible REMF managers are more likely to be chasing similar investment opportunities. Second, the transparency

³ A few examples of earlier work on the effect of scale in mutual fund performance include Tufano and Sevick (1997) who argue cost savings at scale improve performance and Perold and Salomon (1991) who discuss price impact of large transactions driving diseconomies of scale for funds.

⁴ Consistent with this, Chen et al. (2004) and Yan (2008) focus on liquidity constraints (price impact) as the mechanism and show that fund size erodes alpha. Other work on the topic includes Pollet & Wilson (2008), Ferreira et al. (2012), Reuter and Zitzewitz (2021), and Zhu (2018).

of REITs would also suggest a relatively limited number of investment opportunities that REMFs could implement. For example, Hardin et al. (2005) examine stock anomalies in REITs and argue that due to higher transparency, the market for REITs is more efficient.⁵ It follows that a more efficient market would presumably lead to fewer idiosyncratic investment opportunities. Both of these features suggest REMFs would be more likely to be chasing similar strategies than diversified funds.

Although the effect of industry size is unexamined to date in the REMF literature, several papers to date test the relation between fund size and performance for REMFs. The most recent paper to focus on fund level diseconomies of scale is Chou and Hardin (2014) and they find evidence consistent with Berk and Green (2004). However, Pástor et al. (2015) and Zhu (2018) note an endogeneity concern in the relation between fund size and fund alpha that has not been addressed in any previous REMF studies. Whether the findings of Chou and Hardin (2014) hold after controlling for endogeneity is an open empirical question we address in this paper.

Using a sample of REMFs from 1995 to 2019, we examine the effect of industry and fund-level economies of scale on REMF performance. We document a negative relation between industry size and REMF performance consistent with diseconomies of scale at the industry level but not at the fund level. When examining the effect of fund size and industry size in the same regression, industry size continues to be negative and significant and fund size remains insignificant. Overall, we interpret these results as consistent with our hypothesis that industry level diseconomies of scale impact REMF performance. We next perform cross-sectional analyses to test whether some funds are likely to be more sensitive to the industry-level economies of scale than others. We find that the negative relationship between industry size and fund alpha is stronger for firms with higher turnover, expense ratios, and volatility. Each of these represents funds that are more active in identifying the investment opportunities that become more elusive as industry size increases. Our findings further corroborate the industry diseconomies of scale theory.

Our contribution to the existing literature is threefold. First, we are the first to examine industry level economies of scale in the REMF industry. Although a large body of work has been dedicated to understanding REMF performance, we are the first to provide evidence that the size of the industry materially affects performance. Critically, our results provide a plausible explanation for the dispute in the literature as to why earlier papers find REMFs generate alpha and later papers typically do not.

Second, we reexamine the relation between fund size and fund performance in the REMF industry after addressing the endogeneity concerns. The result changes after addressing the endogeneity problem, suggesting the analysis is worthwhile. The fund level diseconomies of scale documented by Chou and Hardin (2014) have significant implications for REMF fund managers. For example, a popular sentiment

⁵ The higher transparency is due to the simplicity of their operations (i.e., buy and operate real estate) along with the fact that they disclose the underlying property portfolios.

in the diversified mutual fund space is that staying small and nimble is important.⁶ If true, REMFs may need to stay small in order to perform well. However, our result that fund size does not hinder performance liberates the funds to grow without concern for their alpha generation ability. Given that funds cannot control overall industry size, their incentive remains to grow as large as possible and maximize assets under management.

Finally, we contribute to the broader diversified mutual fund literature by examining the effect of sector-level industry economies of scale for mutual funds. In REITs, where investment opportunities are narrow, industry-level economies of scale are important. To our knowledge, we are the first to apply the theory of Pástor et al. (2012) to a subset of sector funds. The fact that industry level diseconomies of scale exist at the sector level as well as at the diversified fund level provides further support for the theory.

The remainder of our paper is organized as follows: the second section discusses the related literature, third section details the data and summary statistics, fourth section discusses the empirical methods utilized in the study, fifth section documents the results, sixth section examines the robustness, and seventh section concludes.

Related Literature

The REMF industry began to expand rapidly in the mid-1990s and academic interest followed. Consistent with the broader mutual fund literature, the vast majority of research is focused on the performance of REMFs. Perhaps unsurprisingly, the findings have been mixed regarding whether REMFs are able to consistently outperform their passive benchmarks.

We begin by organizing the REMF literature in Table 1. This table covers papers that examine REMFs and focus on performance or fund size. We highlight the data source, sample period, and primary findings. A few notable observations can be made. First, all papers except one utilize CRSP, Morningstar, or some combination of the two to identify REITs. We utilize both CRSP and Morningstar. Second, only Lantushenko and Nelling (2020) and MacGregor et al. (2020) utilize a sample period that extends beyond 2010. Our sample which extends to 2019 covers a much larger period than most studies to date.

Findings related to firm performance and fund size are notably mixed. Four papers document outperformance, two documents underperformance, and eight document performance that is indistinguishable from the benchmark. It is difficult to establish a consensus in the literature given the widely varying sample periods and empirical methods.

⁶ “The fund manager has to know when to say when, especially in the case of small-cap funds. The fund should stay small and nimble. If it becomes too large, then performance will dwindle.” Steven Roge, Roge Partners Fund (Link: <https://www.thestreet.com/investing/funds/mutual-funds/mutual-fund-bets-on-managers-not-stocks-11015545>).

Table 1 Literature summary on REMF performance and fund size

Paper	Data source	Sample period	Performance	Fund size	Address fund size endogeneity?	Address industry size?
Chou and Hardin (2014)	CRSP	1994–2006	Zero	Negative	No	No
Edward and Daniel (2000)	Morningstar	1996–1998	Zero	Zero	No	No
Kallberg et al. (2000)	Morningstar	1987–1998	Positive	Positive	No	No
Lantushenko and Nelling (2020)	CRSP	1995–2015	Zero	Zero	No	No
Lin and Yung (2004)	WSJ and Yahoo	1993–2001	Negative	Positive	No	No
MacGregor et al. (2020)	CRSP	1992–2016	Zero	Negative	No	No
Kaushik and Pennathur (2012)	CRSP	1990–2008	Positive	Positive	No	No
Hartzell et al. (2010)	CRSP	1994–2005	Zero	Zero	No	No
Derwall et al. (2009)	CRSP	1980–2008	Zero	N/A	N/A	No
Chiang et al. (2008)	Morningstar	1982–2003	Zero	N/A	N/A	No
Cici et al. (2011)	CRSP	1995–2006	Positive	N/A	N/A	No
Gallo et al. (2000)	Morningstar	1991–1997	Positive	N/A	N/A	No
Kaushik and Pennathur (2013)	CRSP	1990–2009	Negative	N/A	N/A	No
Rodriguez (2007)	CRSP/Morningstar	1999–2004	Zero	N/A	N/A	No

This table summarizes the existing literature to date specifically focusing on REMF performance, fund size, or industry size. The data source refers to the primary data source used to identify REMFs (some papers also using Thompson Reuters to obtain fund holdings). Many papers have more than one performance result; we provide the primary finding of the paper

Early work generally pointed to outperformance for REMFs. Specifically, both Kallberg et al. (2000) and Gallo et al. (2000) document the ability of REMFs to generate positive alpha. However, contrary findings have emerged through time as more data and different benchmark models are proposed. Lin and Yung (2004) argue that the documented outperformance of REMFs is due to the outperformance of the real estate sector. When they include the real estate index in the benchmark model, they find no significant alpha for REMFs. Although many studies have since found no evidence of outperformance (Chiang et al., 2008; Derwall et al., 2009; Hartzell et al., 2010; Kaushik & Pennathur, 2013; MacGregor et al., 2020), others document evidence of outperformance in certain circumstances (Cici et al. 2011; Kaushik & Pennathur, 2012; Chou & Hardin, 2014).

Many studies that document insignificant alphas focus on the utilization of the correct benchmark for evaluating REMFs. Chiang et al. (2008) first document positive alpha for REMFs using the capital asset pricing model and Fama and French's (1993) three-factor model. However, after they augment the benchmark models to adjust for the fact that REMFs invest predominantly in REITs, the outperformance vanishes. Derwall et al. (2009) show that although REMFs exhibit significant alpha using traditional asset pricing models, when REIT momentum is accounted for, alpha disappears. Similarly, Hartzell et al. (2010) find that a greater percentage of REIT funds exhibit outperformance than random chance would predict when using a single-index benchmark model. However, with a more complex multiple-index model, the number of REIT funds exhibiting outperformance drops from 6 to 0.7%.

The recent studies that identify some evidence of outperformance typically have some caveats or use alternative measurements of performance. For example, Chou and Hardin (2014) document a positive and significant gross alpha, but net alpha is indistinguishable from zero. Kaushik and Pennathur (2012) document outperformance but only after excluding the real estate driven financial crisis of 2007–2008. Cici et al. (2011) focus on an alternative measure of performance. Rather than focus on fund-level analysis, they focus on the actual fund holdings and trading activities of the REMF. This allows them to isolate whether REMF managers are able to select outperforming REITs. They document superior REMF REIT-selection ability. This result holds even after benchmarking the REITs against other REITs with similar characteristics such as property type.⁷

Rather than testing whether REMFs exhibit outperformance as a group, the focus of our paper is on fund and industry level determinants of alpha. Although the investigation of an industry level determinant of alpha for REMFs is novel, several studies have examined various fund level determinants. Examples of variables investigated include fund size, expense ratio, turnover, fund age, past performance, and various fund fees.⁸ The general consensus of the literature is markedly mixed on each of these determinants. One

⁷ Ro and Gallimore (2014) also focus on fund holdings to examine herding and momentum trading. They find that neither are particularly productive trading strategies for REMFs.

⁸ Another notable investigation of fund level determinants on performance is Lantushenko and Nelling (2020) who examine activeness measures and find no consistent evidence that REMF activeness impacts alpha.

example is the findings to date on the relation between fund age and fund performance. In our analysis of the literature, we note that two studies find a positive relation between performance and fund age (Kallberg et al., 2000; Chou & Hardin, 2014), two studies find a negative relation (Edward & Daniel, 2000; Lantushenko & Nelling, 2019), and two studies find no relation (Lun & Yung, 2004; MacGregor et al., 2020).

A key variable of interest in our study is fund size. As shown in Table 1, three papers document a positive relation with performance, two document a negative relation, and three document no relation.⁹ While several papers include fund size as a control variable or additional test, Kallberg et al. (2000) and Chou and Hardin (2014) study fund size as a primary variable of interest and find opposite results. There are competing theories for whether the relation between fund size and fund performance should be positive or negative for REMFs. A positive relation suggests economies of scale. Larger funds can afford greater costs of information and potentially reduce transactions costs (Kallberg et al., 2000). A negative relation suggests diseconomies of scale. Larger funds face liquidity constraints and are unable to capitalize on investment opportunities due to price impact and a limited number of investment opportunities (Chou & Hardin, 2014).

We draw from the broader mutual fund literature's examination of returns to scale for both empirical methods and theoretical support. The theory underpinning fund level returns to scale is from Berk and Green (2004). They argue that capital will flow to funds that outperform and from funds that underperform. Because of diseconomies of scale, as outperforming funds receive additional capital, their outperformance will not persist. Chen et al. (2004) and Yan (2008) document empirically that the negative returns to scale are driven by liquidity concerns. Specifically, Chen et al. (2004) document that the negative relation between fund size and performance is more pronounced among funds that have to invest in small and illiquid stocks. Similarly, Yan (2008) documents that the negative relation is concentrated in funds with less liquid portfolios, growth funds, and those with high turnover.

In contrast, Pástor et al. (2015) argue that past empirical examinations of fund size on performance suffer from an endogeneity problem. They argue that fund size is not randomly assigned. More skilled managers are likely to manage larger funds. Given skill is likely correlated with both size and performance and skill is unobservable, the estimation of fund size on fund performance suffers from an omitted variables bias. One solution is to add fund fixed effects, however, this creates another bias that results from a positive correlation between fund size changes and unexpected fund returns (Pástor et al., 2015). After controlling for the endogeneity problem, Pástor et al. (2015) document an insignificant relation between fund size and fund performance.¹⁰ However, Zhu (2018) challenges the empirical method of Pástor et al. (2015). After updating the empirical method, Zhu (2018) documents

⁹ Kallberg et al. (2000), Lin and Yung (2004), and Kaushik and Pennathur (2012) document a positive relation. Chou and Hardin (2014) and MacGregor et al. (2020) document a negative relation. Edward and Daniel (2000), Lantushenko and Nelling (2019), Hartzell et al. (2010) document no relation.

¹⁰ Zhu (2018) challenge the empirical method of Pástor et al. (2015) and after altering the method find a negative relation between fund size and fund performance. We utilize the model of both Pástor et al. (2015) and Zhu (2018) for robustness.

a negative relation between fund size and fund alpha. We discuss the Zhu (2018) method in the next section as we use it for our primary analysis.

Although fund level returns to scale have been examined dating back to the early 2000s, Pástor et al. (2012, 2015) are the first to examine industry returns to scale. They argue each managers ability to beat the market declines as the active management industry grows (Pástor & Stambaugh, 2012). Empirically, they document evidence consistent with diseconomies of scale for mutual fund industry size (Pástor et al., 2015). We extend their work by examining the role of fund level and industry level returns to scale in the REMF industry.

Data

We obtain monthly fund return data from the CRSP (Center for Research in Security Prices) Survivor-Bias-Free US Mutual Fund Database for the period between January 1995 and December 2019. This sample period begins following the passage of the 1993 Revenue Reconciliation Act that led to explosive growth in the industry. Our sample begins in 1995 to be consistent with other recent studies on REMFs including Lantushenko and Nelling (2020), Hartzell et al. (2010), Chou and Hardin (2014), and Cici et al. (2011).

We apply a strict set of screens to ensure we are only capturing domestic REMFs that invest primarily in public listed REITs. We keep funds with Lipper objective code “RE”, Lipper class code “RE”, Strategic Insight code “RLE” or CRSP objective code “ESDR”. Funds where none of the Lipper or Strategic Insight codes were available are excluded from the analysis. We also search for fund names that contained “real estate”, “REIT”, or “Realty.” Our focus is on domestic REMFs, so we excluded funds that contained “international”, “emerging market”, “global”, “euro”, “currency”, or “world” in their fund name. We are focused on open-ended actively managed funds. Therefore, we also screened out funds flagged as index funds or ETFs using the “index_fund_flag” and “et_flag” variables in CRSP. Our focus is on REMFs that invest primarily in REITs. Therefore, we dropped funds that appeared to be real estate fixed income funds. We reviewed the final list of funds manually to screen out any leftover funds that are not REMFs. This yielded us a total of 92 unique funds. We then searched these 92 funds in Morningstar Direct to cross-check their Morningstar categories. We found that five of the 92 funds were not identified as U.S. Real Estate funds by Morningstar. Hence, we removed these five funds which gave us a final sample of 87 unique open-ended domestic REMFs.

In order to ensure that the funds in our sample are mostly invested in public listed REITs, we also examine the fund holdings. This is particularly important because our industry size variable assumes that REITs make up the investible universe for REMFs. To the extent REMFs have significant holdings outside of public listed REITs, our measure is an imperfect proxy for industry size. For this purpose, we used the Thomson Reuters Mutual Fund Holdings Database and linked it to the CRSP mutual fund database using MFLINKS. We found that funds in our sample, on average, invest 90.74% of their total net assets in public listed REITs. This finding is almost identical to that of Ro and Gallimore (2014). Although the

measure of industry size contains error given REMFs may invest in other stocks, the vast majority of their holdings are public listed REITs.

The CRSP database also contains information on fund characteristics such as total net assets, expense ratio, and turnover. We use the MFLINKS database to combine multiple share classes into fund-level portfolios. For these funds, we also compute returns, expenses, turnover ratio, and other characteristics by taking an average weighted by most recent total net assets of a share class. For total net assets, we sum them. Lastly, following Elton et al. (1996) and Evans (2010), we include two additional screens to address the potential of incubation bias. First, we eliminate funds with total net assets less than \$15 million. Second, we delete any observations that precede the offer date in the CRSP database.

Our primary variables of interest are *Gross alpha*, *Fund size*, and *Industry size*. Following Pástor et al. (2015) and Zhu (2018), we define *Gross alpha* as the fund return including all expenses minus the benchmark return. Here, the benchmark is obtained using the Morningstar Category Benchmarks. For all the funds in our sample, the Morningstar Category Benchmark is the S&P United States REIT Index. We obtain the returns for this index from Bloomberg. Our sample starts in 1995 and the S&P United States REIT Index is only available starting January 2001. Therefore, for the pre-2001 analysis, we use the NAREIT index instead. From 2001 to 2019, the correlation between the monthly returns on NAREIT index and the S&P United States REIT Index is 99.77%. Additionally, the results in this paper similar if we only use the S&P United States REIT Index and limit our sample to start in January 2001.

Fund size is the total net assets under management at the end of the previous month. The value is adjusted for inflation based on 2011 as the base year following Pástor et al. (2015). *Industry size* is the sum of all REMFs fund sizes scaled by the market capitalization of all public listed REITs as reported by NAREIT as of the end of the previous month. The various control variables used throughout the paper are defined in the [Appendix](#). We exclude fund-month observations with missing data. Our final sample consists of 87 funds and 12,480 fund-month observations.

Table 2 displays summary statistics for the REMF funds. The mean gross return is 1.04% per month for the typical fund. However, gross alpha is merely 0.14%. The

Table 2 Summary statistics

Variables	N	Mean	Std	25th	50th	75th
<i>Gross return</i>	12,480	0.0104	0.0573	-0.0163	0.0146	0.0421
<i>Net return</i>	12,480	0.0094	0.0573	-0.0172	0.0136	0.0411
<i>Gross alpha</i>	12,480	0.0014	0.0161	-0.0030	0.0017	0.0062
<i>Fund size</i>	12,480	760.96	1236.89	86.00	246.95	865.35
<i>Fund age (in months)</i>	12,480	152.98	87.44	84.00	142.00	209.00
<i>Industry size</i>	12,480	0.1234	0.0320	0.1016	0.1265	0.1432
<i>Turnover ratio</i>	12,480	0.7035	1.0852	0.2600	0.4800	0.8100
<i>Expense ratio</i>	12,480	0.0119	0.0039	0.0096	0.0111	0.0144
<i>Std. Returns</i>	12,480	0.0502	0.0274	0.0363	0.0423	0.0516

This table shows summary statistics for our sample of REMFs for the period 1995 to 2019. All observations are on a fund-month basis. Each variable is defined in the [Appendix](#)

typical fund in our sample has approximately \$760 million in assets under management and has existed for almost 13 years. The average turnover ratio is 0.70 and the average expense ratio is 1.19%. The average industry size coefficient is 0.1234. If REMFs invested 100% of their assets in REITs, this would indicate that REMFs hold 12.34% of the market share of REITs. However, REMFs hold slightly less than 12.34% of the market share of REITs because 90.77% of their assets are invested in REITs and 9.23% are invested elsewhere.

Empirical Methods

In this paper, our goal is to examine economies of scale by studying the impact of lagged fund size and industry size on the performance of a fund. Given some funds are younger than others, we are working with an unbalanced linear panel. Estimating the effect of fund size on the benchmark adjusted performance is challenging because the fund size is determined endogenously with managerial skill. For example, larger funds could also have highly skilled managers. A simple OLS regression where the dependent variable is the benchmark adjusted performance and lagged fund size is the independent variable does not account for the endogenously related skill and suffers from an omitted variable bias. Manager skill is not observable and likely related to both fund size and fund alpha, creating an omitted variable bias.

The omitted variable bias can be mitigated by including fund fixed effects to capture any time invariant, fund specific variance. However, Pástor et al. (2015) argue that although the inclusion of fund fixed effects alleviates the omitted variable bias previously discussed, it introduces a different type of bias. Fund size generally increases following higher returns and this creates a mechanical positive contemporaneous correlation between changes in fund size and unexpected fund returns. Fixed effects estimators use demeaning processes such as either subtracting the full time-series mean from each variable or taking differences in the variable values in consecutive periods. Pástor et al. (2015) and Zhu (2018) point out that because of the contemporaneous correlation, the demeaning processes induce a bias in the OLS fixed effect estimator.

To address this problem, they propose using use a recursive demeaning two-stage regression proposed by Moon and Phillips (2000). In this approach, they propose regressing forward demeaned benchmark adjusted performance on forward demeaned fund size in the second stage. Additionally, in the first stage regression, they use the backward demeaned fund size as an instrument for the forward demeaned fund size. This approach may not address for possible heteroscedasticity and autocorrelation in the error terms. Therefore, we use Newey & West (1987) standard error correction.

Following Pástor et al. (2015), Zhu (2018) suggest an improvement on the empirical method. Specifically, Pástor et al. (2015) impose a zero intercept in both the first stage and second stage regressions. Zhu (2018) shows that imposing zero intercept in the first stage regression reduces the first stage R-square and hence, the efficiency of the estimate.

To test whether the Zhu (2018) adjustment is necessary in our setting, we run the first stage regression using both methods with REMF data. The results are in Table 3. The difference between the first stage fit for the two methods is quite apparent. In this table, the first two columns show the first stage regression coefficients and R-squares using the Pástor et al. (2015) method and the last two columns show the first stage regression coefficients and R-squares using the Zhu (2018) method. Column 1 and column 3 contain the results where the only independent variable is the forward demeaned fund size. Column 2 and column 4 show the results for the first stage regression where both the forward demeaned fund size and forward demeaned industry size are the independent variables. Following Pástor et al. (2015) and Zhu (2018), we assume that the industry size is exogenous and instrument for the forward demeaned fund size using the backward demeaned fund size. Columns 1 and 2 show that the coefficient on the instrument in the first stage regression is significant, although the R-square is modest. However, in Columns 3 and 4, where we follow Zhu (2018), the fit is drastically improved. The R-square coefficient increases from 4.53% in Column 1 to 58.34% in column three. This evidence suggests the Zhu (2018) method is a better instrumental regression in assessing the impact of the fund size on fund performance.

Throughout the paper, we follow Zhu (2018) and use the recursive demeaned estimator of fund size in our tests to account for all the biases discovered by past literature. One of the contributions of this paper is to investigate whether the negative relationship between fund size and returns established by previous studies holds after controlling for endogeneity. To be exhaustive, in our sixth section we show that our results are robust to the Pástor et al. (2015) method.

Table 3 Comparison of recursive demeaning methods

Variables	Pástor et al. (2015) Method		Zhu (2018) Method	
	(1)	(2)	(3)	(4)
<i>Intercept</i>			-550.65	-562.97
<i>T-stat</i>			(-43.40)	(-44.46)
<i>Instrument</i>	0.2090	0.2052	0.4747	0.4753
<i>T-stat</i>	(24.33)	(23.86)	(132.18)	(133.04)
<i>N</i>	12,480	12,480	12,480	12,480
<i>First Stage R²</i>	4.53	4.83	58.34	58.78

This table reports the results of first stage regressions for both the Pástor et al. (2015) and Zhu (2018) recursive demeaning methods. The first stage intercept is constrained to zero in the Pástor et al. (2015) method. In all columns, *Gross alpha* is the dependent variable. Instrument is the backward demeaned fund size. R^2 is in percent. For the regressions in column 1 and column 3 the independent variables consist of only the *Fund size*. For the regressions in column 2 and column 4 the independent variables are the *Fund size* and the *Industry size*, and *Industry size* is assumed to be exogenous

Our primary results suggest diseconomies of scale at the industry level is the driver of eroding alpha. The underlying theory is motivated by liquidity and price impact concerns, so we further analyze whether specific fund characteristics of interest affect the relation between *industry size* and performance. Particularly, we look at REMFs with high expense ratios, turnover ratios, and return volatility. We use these variables because each of them proxy for subsets of funds that are most likely to be impacted by limited investment opportunities. Effectively, these funds are more active and riskier. We sort the funds by variable of interest, then add to our model a variable capturing the funds in the top quartile, then interact it with industry size:

$$\begin{aligned} \text{Gross } \alpha_{it} = & \alpha_i + \beta_1 \text{Industry size}_{it-1} + \beta_2 \text{Fund Characteristics top } Q_{it-1} \\ & + \beta_3 \text{Industry size}_{it-1} * \text{Fund Characteristic} * \text{top } Q_{it-1} + \varepsilon_{it} \end{aligned} \quad (1)$$

Where *Fund Characteristic top Q* is an indicator variable equal to one if the fund's characteristic of interest is in the top quartile at the end of period t-1, zero otherwise. We next detail the data and summary statistics.

Results

We begin our analysis by examining industry size and fund size over time. Figure 1 graphs industry size over time. Interestingly, we observe significant time-series variation, and the variable of interest is not constantly climbing. A sharp increase

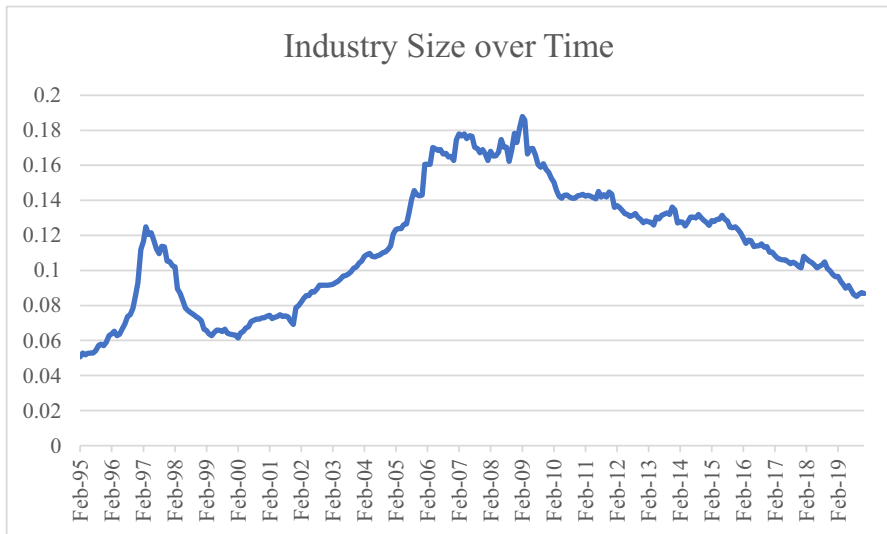


Fig. 1 Shows the *Industry size*, defined as the sum of funds' assets under management (AUM) divided by the total market value of REIT stocks over time. It represents the fraction of the total REIT stock market capitalization owned by REMFs included in the sample at the end of the month

is observed in the mid-1990s followed by an equally sharp decline into 2000. The variable more than doubles from 0.07 to 0.18 from 2000 to 2009 and has steadily declined in the last decade of our sample. As of the end of 2019, the value was approximately 0.09.

Why does the variable exhibit such variation in an industry that has grown fairly consistently in assets under management and average fund size? The industry size variable has two components: the total assets under management for all REMFs in a given month scaled by the total market capitalization of REITs. Industry size is scaled by REIT market capitalization because we are concerned with how large the REMF industry is *relative* to the investable universe. If the investable universe grows by more than the REMF industry, the industry becomes less crowded. This occurs even if the REMF industry grows via more funds being created or existing funds increasing assets under management.

To understand the time-series variation in the industry size variable, it is useful to examine the two components separately. Figure 2 breaks out the total REMF assets under management and the total REIT market capitalization. We now observe the expected trend that both REMF assets and REIT market capitalization have grown significantly throughout the sample. The variation in the industry size variable represent times when the two trends deviate from one another.

Interestingly, in the period leading up to the financial crisis, REMFs exhibited more rapid growth than the REIT market capitalization. During the ensuing crash, REMFs exhibited a stronger decline than the REIT market capitalization. Another notable deviation explains why the industry size variable has declined in recent years. Although the REIT market capitalization has seen major growth from the trough of the financial crisis through 2019, assets under management for REMFs has been less dramatic. One plausible explanation is the shift from active management to passive management in the last decade. If an investor shifts his capital from a REMF into a passive REIT index, this negatively impacts the assets under management but should not impact REIT market capitalization. If the industry level diseconomies of scale story holds true, a continued trend from active to passive management could produce opportunities for REMFs in the future.

We next examine the average fund size over time and the number of REMFs. Figure 3 displays the average fund size over time and Fig. 4 displays the number of funds. Although the average fund size has grown over time, the number of funds has stagnated since the financial crisis. This suggests that the majority of the growth in total assets under management following the financial crisis stems from funds increasing in size rather than more new funds entering than old ones exiting. Overall, this is consistent with a muted appetite for active management relative to the growth of the underlying REIT industry.

Table 4 presents our primary regression analysis. We regress industry size and fund size on gross alpha using various regression specifications. In columns 1–3, we use standard ordinary least squares without fixed effects. Column 1 examines fund size alone, column 2 examines industry size alone, and Column 3 examines them together. Fund size is negative, suggesting diseconomies of scale at the fund

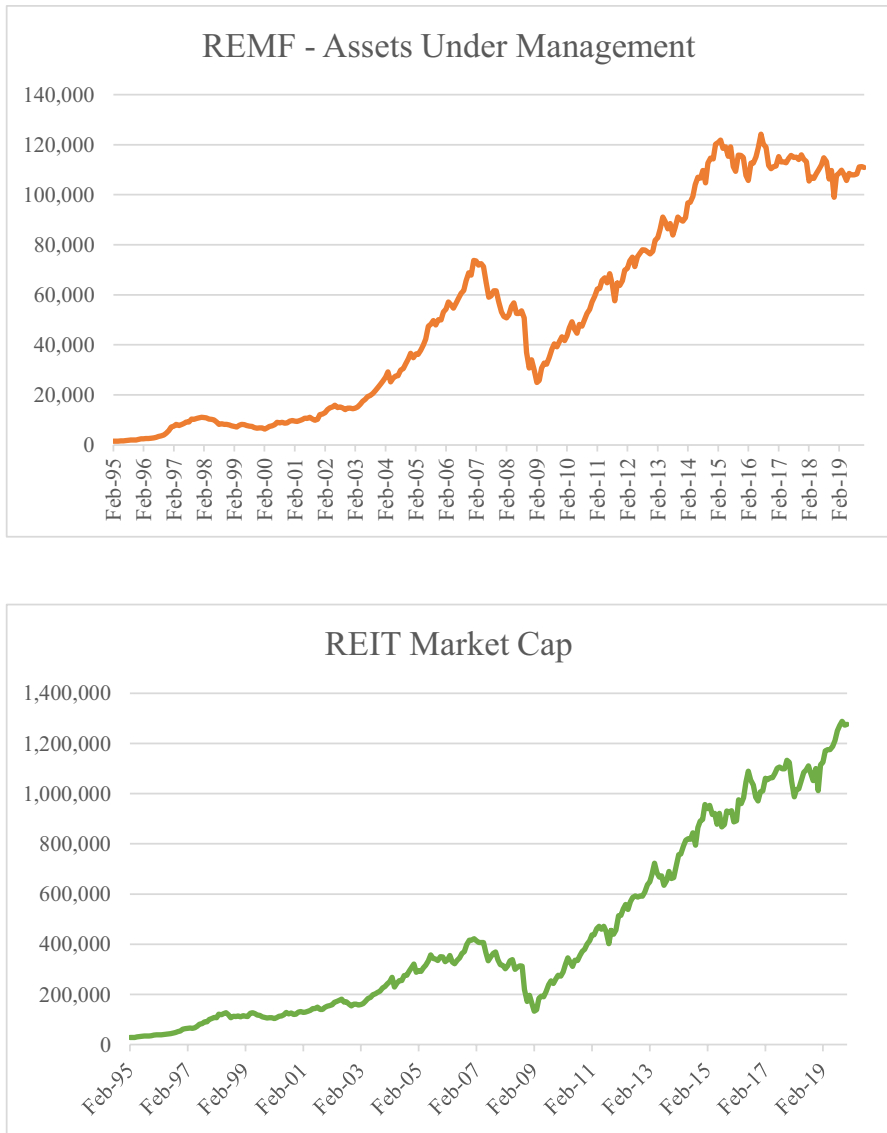


Fig. 2 Shows the industry size breakdown. The top panel of Fig. 2 shows the *assets under management* (AUM) of the REMFs in the sample at the end of the month in \$ million, over time. The bottom panel shows the total market capitalization of the REIT industry at the end of the month in \$ million, over time

level, but it is statistically insignificant. In contrast, industry size is negative and significant beyond the 10% level, both alone and in the specification with fund size included.



Fig. 3 Shows the size of the average REMF in the sample, measured by the fund's assets under management in millions at the end of each month

However, the ordinary least square regressions present an omitted variables bias. Therefore, we next consider fund fixed effects to mitigate the concern that skill is an omitted variable. Columns 4–6 examine the effect of fund and industry size on alpha. The industry size variable continues to be negative and increases in both economic magnitude and statistical significance. Interestingly, fund size is now negative

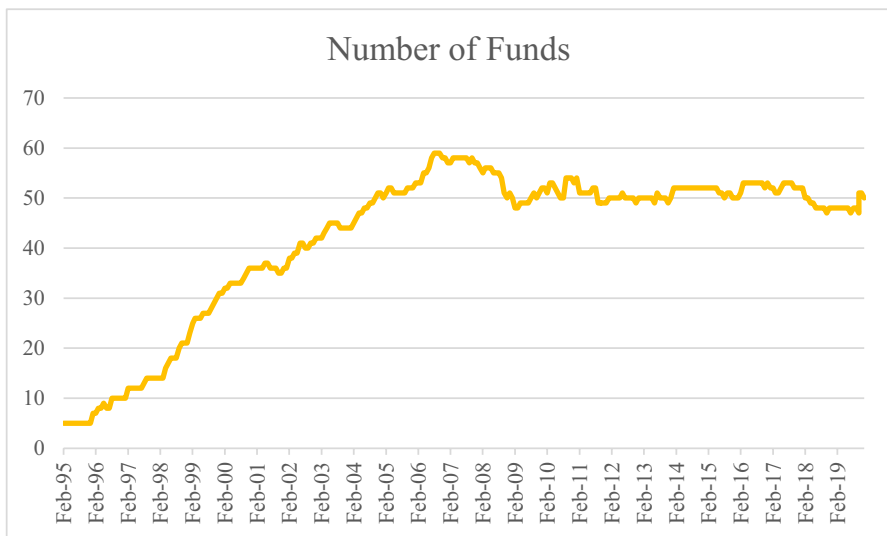


Fig. 4 Shows the number of individual REMFs over time for each month in our sample

Table 4 Relation between fund size and industry size and performance

Variables	OLS no FE			OLS with FE			Recursive demeaning		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
<i>Intercept</i>	0.0014	0.0024	0.0024						
<i>T-stat</i>	(8.14)	(4.13)	(4.11)						
<i>Industry size</i>	-0.0083		-0.0083	-0.0113		-0.0106	-0.0120		-0.0115
<i>T-stat</i>	(-1.83)		(-1.84)	(-2.28)		(-2.14)	(-2.37)		(-2.26)
<i>Fund size</i>		-0.0099	-0.0122		-0.1621	-0.1511		-0.1144	-0.1135
<i>T-stat</i>		(-0.21)	(-0.26)		(-2.11)	(-1.97)		(-1.12)	(-1.11)
<i>N</i>	12,480	12,480	12,480	12,480	12,480	12,480	12,480	12,480	12,480
<i>R</i> ²	0.03	0.00	0.03	0.55	0.55	0.58	0.05	0.01	0.05

This table reports our base results of the impact of *Fund size* and *Industry size* on *Gross alpha* using three different estimations. The first model in columns 1 to 3 is a classic ordinary least squares regression with no fixed effects. The second model in columns 4 to 6 is an ordinary least square regression that includes fund fixed effects. The third model in columns 7 to 9 follows the Zhu (2018) method using a recursive demeaned estimator. The table reports the coefficients of three specifications per model, one with each variable of interest separately and another with the two variables in the same specification. R^2 is in percent. The coefficients on *Fund size* have been multiplied by one million for readability. Each variable is defined in the [Appendix](#)

and statistically significant beyond the 5% level. The coefficients on fund size have been multiplied by one million for readability since fund size is a dollar value being regressed on alpha. This finding mirrors the primary results of Chou and Hardin (2014), who argue for diseconomies of scale at the fund level.

As Pástor et al. (2015) and Zhu (2018) note, solving one issue of endogeneity introduces another. Our final specification utilizes the recursive demeaning procedure of Zhu (2018) to address these concerns. Columns 7–9 present the results. *Fund size* returns to an insignificant negative value with t-statistics -1.12 and below. This finding is significant as it overturns a key result in the literature to date. After controlling for endogeneity, we no longer find that there exist diseconomies of scale at the fund level. Equally important, *Industry size* continues to be negative and statistically significant. Notably, across all specifications, with and without fund size included in the regression, industry size coefficients and t-statistics are quite stable. The economic significance of the industry size coefficient is notable. A one standard deviation increase in the industry size variable is associated with a 3.7 basis point decrease in monthly gross alpha. This is equivalent to approximately 44 basis points per year. We interpret these results as consistent with strong evidence that industry size is an economically and statistically important driver of REMF performance, while fund size is not.

Our next set of tests seeks to further understand how industry size impacts REMF alpha. Although industry size exhibits increases and decreases throughout the sample, it does trend upward over time on average. We include a time trend variable, equal simply to the number of months from the beginning of the sample period and include it in the regression to mitigate the concern that *Industry size* is simply capturing a downward trend in alpha over time. Additionally, total assets under

management is computed as the number of funds multiplied by the average fund size. We examine these separately and together to see if one aspect of industry size appears to drive the result.

Table 5 presents the results of this test. We utilize the recursive demeaning procedure of Zhu (2018) for each result in this table. Columns 1 to 4 examine each of the aforementioned variables individually and column 5 combines them. The time trend and average fund size are insignificant whereas the number of funds is marginally positively significant in the full model of column 5. This is consistent with our expectations given the industry and fund size results of Table 2. Critically, industry size remains negative and statistically significant after controlling for these variables. This alleviates the concern that a time trend is driving the relation between industry size and alpha.

The industry-level diseconomies of scale theory proposed by Pástor and Stambaugh (2012) is motivated by liquidity and price impact concerns. If indeed that is the case, we expect some funds to be more impacted by an overcrowding industry than others. To test for this, we identify funds that are in the top quartile of turnover ratio, expense ratio, and the standard deviation of returns. Each of these represents funds that are more active in their trading. Funds that rely heavier on identifying investment opportunities will exhibit steeper diseconomies to scale as the opportunities dwindle. We interact industry size with indicators for each of these variables.

Table 6 presents the results of these cross-sectional regressions. Across all specifications, the industry size variable remains negative and significant. This suggests that the effect of industry size on alpha is widespread and not confined to the subset of firms identified by the indicator variables. However, for each interaction, the coefficient is negative and significant as well. These results suggest that industry size erodes alpha for the broad universe of REMF funds, but funds with high turnover ratios, expense ratios, and volatility of returns are particularly

Table 5 Impact of fund and industry size over time

Variables	(1)	(2)	(3)	(4)	(5)
<i>Industry size</i>	-0.0120				-0.0272
<i>T-stat</i>	(-2.37)				(-2.28)
<i>Average Fund size</i>		-0.0876			0.7762
<i>T-stat</i>		(-0.26)			(0.36)
<i>Number of funds</i>			-172.73		97.48
<i>T-stat</i>			(-1.63)		(1.80)
<i>Time trend</i>				-2.04	-11.36
<i>T-stat</i>				(-0.85)	(-0.69)
<i>Fund F.E.</i>	Yes	Yes	Yes	Yes	Yes
<i>N</i>	12,480	12,480	12,480	12,480	12,480
<i>R²</i>	0.05	0.00	0.02	0.01	0.08

This table reports the coefficients of the impact of several variables of interest on the dependent variable *Gross alpha* using recursive demeaned estimator. R^2 is in percent. The coefficients on *Fund size* have been multiplied by one million for readability. Each variable is defined in the [Appendix](#)

Table 6 The impact of fund characteristics on the industry size and performance relation

Variables	(1)	(2)	(3)	(4)
<i>Industry size</i>	-0.0120	-0.0333	-0.0711	-0.0375
<i>T-stat</i>	(-2.37)	(-2.75)	(-4.98)	(-3.80)
<i>High Turnover ratio</i>		0.0816		
<i>T-stat</i>		(1.95)		
<i>Industry size * High Turnover</i>		-0.0069		
<i>T-stat</i>		(-2.16)		
<i>High Expense ratio</i>			0.2600	
<i>T-stat</i>			(4.70)	
<i>Industry size * High Expense ratio</i>			-0.0330	
<i>T-stat</i>			(-4.35)	
<i>High Std returns</i>				0.0959
<i>T-stat</i>				(3.34)
<i>Industry size * High std. returns</i>				-0.0135
<i>T-stat</i>				(-3.48)
<i>N</i>	12,480	12,480	12,480	12,480
<i>R</i> ²	0.05	0.16	0.22	0.13

This table reports the coefficients of a series of specifications where specific deal characteristics are interacted with Industry size as follows:

$$\text{Alpha}_{it} = \alpha_i + \beta_1 \text{Industry size}_{it-1} + \beta_2 \text{HighFChar}_{it-1} + \beta_3 \text{Industry size}_{it-1} * \text{HighFChar}_{it-1} + \epsilon_{it}.$$

FChar represents one of the following fund characteristics: *High turnover ratio*, *High expense ratio*, and *High std. returns*. *High turnover ratio* is an indicator variable equal to one if the fund ranks in the top quartile in terms of turnover ratio, and zero otherwise. *High expense ratio* is an indicator variable equal to one if the fund ranks in the top quartile in expenses, and zero otherwise. *High std. returns* is an indicator variable equal to one if the fund's standard deviation of returns ranks in the top quartile, and zero otherwise. *R*² is in percent. Each variable is defined in the [Appendix](#)

impacted by the size of the industry. The last portion of our analysis produces additional robustness tests.

Robustness

We next produce several robustness tests to ensure our results are consistent. Our tests focus on three dimensions of potential concern: the empirical method, extreme events, and omitted variables.

In Table 3 we provided evidence that the method of Zhu (2018) is more appropriate than the original Pástor et al. (2015) model. However, to alleviate the concern that the results are driven solely by the choice of empirical method, we employ the Pástor et al. (2015) model and rerun the fund size and industry size test. The results are tabulated in Table 7, Panel A. We observe quite similar results to our baseline recursive demeaning regressions in Table 4. *Industry size* continues to be negative, statistically significant, and of similar economic magnitude. *Fund*

Table 7 Empirical method and financial crisis robustness

Panel A: Pástor et al. (2015) recursive demeaning adjustment			
<i>Industry size</i>	(1)	(2)	(3)
<i>T-stat</i>	-0.0120 (-2.37)		-0.0112 (-2.10)
<i>Fund size</i>		-0.2317 (-0.62)	-0.1739 (-0.46)
<i>T-stat</i>		12,480	12,480
<i>N</i>		0.05	0.05
<i>R</i> ²			
Panel B: Excluding the financial crisis			
<i>Industry size</i>	(1)	(2)	(3)
<i>T-stat</i>	-0.0162 (-3.27)		-0.0161 (-3.23)
<i>Fund size</i>		-0.0612 (-0.22)	-0.0984 (-0.36)
<i>T-stat</i>		10,543	10,543
<i>N</i>		0.10	0.10
<i>R</i> ²			

In *Panel A* reports the coefficients of a regression replicating our base results on the impact of *Fund size* and *Industry size* on *Gross alpha* using the recursive demeaning procedure proposed by Pástor et al. (2015). *Panel B* reports the coefficients of a regression replicating our base results excluding observations occurring during the financial crisis (January 2007 to December 2009). *R*² is in percent. Each variable is defined in the [Appendix](#)

size also continues to be negative and statistically insignificant. The T-statistics drop from -1.11 in column 9 of Table 4 to -0.46 in column 3 of Table 7. Therefore, our primary findings are robust to the choice of empirical specification.

A cursory glance at the figures examining assets under management, REIT industry size, fund size, and the number of funds over time display a significant disruption of their typically increasing trend during the financial crisis. Given that the *industry size* is a function of both assets under management and REIT industry size, the parallel downtrend results in less extreme movement of the ratio. However, volatility is still evident in both the *industry size* and *fund size* variables. To alleviate the concern our results may be driven by this major event that originated in the real estate space, we rerun our baseline regressions removing all observations between January 2007 and December 2009. The results are displayed in Table 7, Panel B. The results continue to be consistent. The *industry size* coefficient is negative and statistically significant and *fund size* is insignificant. In fact, the coefficient on *industry size* increases from 0.0115 in Table 4, column 9 to -0.0161 in Table 7, Panel B, column 3. Similarly, the T-statistic increases significantly. Our results are robust to the exclusion of the Global Financial Crisis time period.

Our regression specification follows the work of Pástor et al. (2015) and Zhu (2018). They do not include other control variables in their regressions because their focus is on economies of scale and fund performance. However, one may be concerned that observable fund characteristics are driving the relation between *industry size* and performance. That is, there exists an omitted variables problem. To mitigate this concern, we include fund characteristics as control variables in our baseline regressions. Specifically, we include *turnover ratio*, *expense ratio*, *fund age*, and *std. ret.* The results are produced in Table 8. Interestingly, *industry size* continues to be negative, but statistical significance increases relative to the baseline regressions in Table 4. *Fund size* continues to be negative and statistically insignificant across all specifications.

The full model regression in Table 8 allows for a comparison of cross-sectional drivers of fund performance relative to other REMF studies. We document three consistent results across each of the three columns. First, *expense ratio* is negatively related to *gross alpha*. This result suggests investors would be wise to minimize fees when selecting amongst REMFs. This finding is also consistent with the existing REMF literature, even across notable differences in model specification and sample period. While Chou and Hardin (2014) and Edward and Daniel (2000) document negative but insignificant coefficients of expense ratio on alpha, Lantushenko and Nelling (2020) document a persistently negative relation.

Next, we document a negative relation between *fund age* and *gross alpha*. The existing REMF literature finds inconsistent results regarding *fund age*. Chou and Hardin (2014) and Kallberg et al. (2000) document a positive relation, Edward and Daniel (2000) document a negative relation, and Lantushenko and Nelling (2020), Lin and Yung (2004), and MacGregor et al. (2020) document no relation. Finally, we document a positive relation between a measure of fund risk, the standard deviation of fund returns, and *gross alpha*. We are the first to our knowledge to examine this measure in the REMF literature, and the finding is intuitive. In an efficient market, a fund with higher levels of risk should be commensurate with greater benchmark adjusted performance.

Table 8 Fund characteristics

Variables	(1)	(2)	(3)
Industry size	-0.0192		-0.0181
<i>T-stat</i>	(-3.40)		(-2.81)
<i>Fund size</i>		-0.4470	-0.1300
<i>T-stat</i>		(-1.34)	(-0.35)
<i>Turnover ratio</i>	-0.1161	-0.2195	-0.1467
<i>T-stat</i>	(-0.26)	(-0.48)	(-0.32)
<i>Expense ratio</i>	-1.1117	-0.8063	-1.0861
<i>T-stat</i>	(-3.03)	(-2.25)	(-2.90)
<i>Firm Age</i>	-0.0154	-0.0135	-0.0158
<i>T-stat</i>	(-3.21)	(-2.73)	(-3.23)
<i>Std. returns</i>	0.0198	0.0103	0.0193
<i>T-stat</i>	(2.89)	(1.56)	(2.75)
<i>N</i>	12,480	12,480	12,480
<i>R</i> ²	0.17	0.08	0.17

This table reports the coefficients of a regression that controls for fund characteristics. *Gross alpha* is the dependent variable. *R*² is in percent. The coefficients on *Fund size* have been multiplied by one million for readability. *Turnover ratio* and *Fund Age* coefficients are multiplied by one thousand for readability

Overall, these additional tests point to a robust negative relation between *Industry size* and fund performance. Similarly, the result regarding *Fund size* is also robust. Across all specifications that control for endogeneity, it is statistically indistinguishable from zero. These findings suggest that diseconomies of scale exist at the industry level, rather than the fund level, for REMFs.

Conclusion

A large number of studies in the diversified mutual fund literature examine the performance drivers of active management. Perhaps the most puzzling fact is that active managers do not appear to beat the market on average, yet the industry still exists. Theories by Berk and Green (2004) and Pástor and Stambaugh (2012) rely on rational models to explain this phenomenon. Their theories focus on a competitive allocation of funds and diseconomies of scale at the fund and industry-level.

The goal of our paper is to test these theories in the context of REMFs. We hypothesize that industry diseconomies of scale will be the more important determinant of fund alpha for three reasons. First, the investible universe of REMFs is small and homogenous compared to the broader diversified fund industry and funds are more likely to be chasing similar investment opportunities. As more capital flows into the industry, those opportunities diminish. Second, REITs are known to be more transparent firms than the typical equity. More transparency decreases informational advantages of expert investors. Finally, past research documented outperformance of

REMFs in the infant stages of the industry (Kallberg et al., 2000) and a lack of out-performance as the industry matured (Hartzell et al., 2010).

Consistent with our hypothesis, we find a consistently negative and significant relation between *industry size* and REMF performance. After controlling for endogeneity, we find no relation between *fund size* and REMF performance. This finding is important because it overturns a recent finding of negative returns to scale at the fund level (Chou & Hardin, 2014) and has implications for whether REMF fund managers should stay small to be more effective. The negative relation between industry size and performance is stronger in funds with high turnover ratios, expense ratios, and volatility of returns. We are the first to our knowledge to examine REMF industry level returns to scale. There is much left to be investigated on this topic. For example, one could examine the impact of scale for real estate focused funds in an international setting where there is less competition and less developed REIT industries.

Appendix

Table 9 Variable definitions

Variable Name	Description
Net return	Monthly Net return is obtained from CRSP Mutual Funds – Monthly Returns and Net Asset Values file.
Gross return	Gross return is obtained as net return plus annualized expense ratio divided by 12.
Gross alpha	Following Pástor et al. (2015), Gross alpha for a given fund-month is calculated as the gross fund return (i.e., including expenses and fees) minus the monthly return on the Morningstar Category Benchmark. The Morningstar Category Benchmark for all funds in our sample is the S&P United States REIT Index. This index is available starting in January 2001. Therefore, for pre-2001 period, we use the NAREIT index as the benchmark.
Fund size	Following Pástor et al. (2015), Fund Size is the fund's total net assets under management, inflated to millions of 2011 dollars using the total market capitalization of all stocks in CRSP.
Average Fund Size	For a given month, Average Fund Size is the mean of the fund size (as defined above) of all the funds in our sample.
Industry Size	Following Pástor et al. (2015), Industry size is obtained as the total net assets under management of all REMFs divided by the total market capitalization of all publicly traded REITs in CRSP.
Number of Funds	The Number of Funds for a given month is the amount of unique domestic equity REMFs funds in our sample.
Fund Age	We measure Fund Age as the number of months since the fund's first offer date as listed in the CRSP database.
Turnover Ratio	The minimum of aggregate purchases and sales divided by the average annual asset under management as reported by CRSP.
Expense Ratio	The expense ratio as reported by CRSP. It is the ratio of total investment that shareholders pay for the fund's operating expenses, including 12b-1 fees, waivers, and reimbursements. It is obtained from CRSP and is as of the most recently completed fiscal year.

Variable Name	Description
Std. Returns	The 24-month rolling-window standard deviation of the excess gross returns for a fund.
Time Trend	This variable takes the value one for January, 1990. After that, each month the value of this variable increases by one.

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Declarations

Conflict of interest No conflict of interest exists.

References

- Berk, J. B., & Green, R. C. (2004). Mutual fund flows and performance in rational markets. *Journal of political economy*, 112(6), 1269–1295.
- Chen, J., Hong, H., Huang, M., & Kubik, J. D. (2004). Does fund size erode mutual fund performance? The role of liquidity and organization. *American Economic Review*, 94(5), 1276–1302.
- Chiang, K. C. H., Kozhevnikov, K., Lee, M.-L., & Wisen, C. H. (2008). Further evidence on the performance of funds of funds: the case of real estate mutual funds. *Real Estate Economics*, 36(1), 47–61.
- Chou, W.-H., & Hardin, W. G. (2014). Performance chasing, fund flows and fund size in real estate mutual funds. *The Journal of Real Estate Finance and Economics*, 49(3), 379–412.
- Cici, G., Corgel, J., & Gibson, S. (2011). Can fund managers select outperforming REITs? Examining fund holdings and trades. *Real Estate Economics*, 39(3), 455–486.
- Derwall, J., Huij, J., Brounen, D., & Marquering, W. (2009). REIT momentum and the performance of real estate mutual funds. *Financial Analysts Journal*, 65(5), 24–34.
- Fama, E. F., & French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33 no(1), 3–56.
- Ferreira, M. A., et al. (2012). The flow-performance relationship around the world. *Journal of Banking & Finance*, 36(6), 1759–1780.
- Gallo, J. G., Lockwood, L. J., & Rutherford, R. C. (2000). Asset allocation and the performance of real estate mutual funds. *Real Estate Economics*, 28(1), 165–185.
- Hardin, W. G., Liano, K., & Huang, G.-C. (2005). REIT stock splits and market efficiency. *The Journal of Real Estate Finance and Economics*, 30(3), 297–315.
- Hartzell, J. C., Mühlhofer, T., & Titman, S. D. (2010). Alternative benchmarks for evaluating mutual fund performance. *Real Estate Economics*, 38(1), 121–154.
- Kallberg, J. G., Liu, C. L. & Trzcinka, C. (2000). The value added from investment managers: An examination of funds of REITs. *Journal of Financial and Quantitative Analysis*, 35, 387–408.
- Kaushik, A., & Pennathur, A. K. (2012). An empirical examination of the performance of real estate mutual funds 1990–2008. *Financial Services Review*, 21, 4.
- Kaushik, A., & Pennathur, A. K. (2013). Performance and new money cash flows in real estate mutual funds. *Journal of Financial Research*, 36(4), 453–470.
- Lantushenko, V., & Nelling, E. (2020). Active Management in Real Estate Mutual Funds. *The Journal of Real Estate Finance and Economics*, 61(2), 247–274.
- Lin, C., & Yung, K. (2004). Real estate mutual funds: performance and persistence. *Journal of Real Estate Research*, 26(1), 69–94.
- MacGregor, B. D., Schulz, R., & Zhao, Y. (2020). Performance and market maturity in mutual funds: Is real estate different? *The Journal of Real Estate Finance and Economics*, 63(3), 437–492.
- Moon, H. R., & Phillips, P. C. B. (2000). Estimation of autoregressive roots near unity using panel data. *Econometric Theory*, 16(6), 927–997.

- Newey, W. K., & West, K. D. (1987). Hypothesis testing with efficient method of moments estimation. *International Economic Review*, 28(3), 777–787.
- Pástor, L., & Stambaugh, R. F. (2012). On the size of the active management industry. *Journal of Political Economy*, 120(4), 740–781.
- Pástor, L., Stambaugh, R. F., & Taylor, L. A. (2015). Scale and skill in active management. *Journal of Financial Economics*, 116(1), 23–45.
- Perold, A. F., & Salomon, R. S., Jr. (1991). The right amount of assets under management. *Financial Analysts Journal*, 47(3), 31–39.
- Pollet, J. M., & Wilson, M. (2008). How does size affect mutual fund behavior? *The Journal of Finance*, 63(6), 2941–2969.
- Reuter, J., & Zitzewitz, E. (2021). How much does size erode mutual fund performance? A regression discontinuity approach. *Review of Finance*, 25(5), 1395–1432.
- Ro, S. H., & Gallimore, P. (2014). Real estate mutual funds: Herding, momentum trading and performance. *Real Estate Economics*, 42(1), 190–222.
- Tufano, P., & Sevick, M. (1997). Board structure and fee-setting in the US mutual fund industry. *Journal of Financial Economics*, 46(3), 321–355.
- Yan, X. (2008). Liquidity, investment style, and the relation between fund size and fund performance. *Journal of Financial and Quantitative Analysis*, 43, 741–767.
- Zhu, M. (2018). Informative fund size, managerial skill, and investor rationality. *Journal of Financial Economics*, 130(1), 114–134.
- Edward, S., & Daniel, E. (2000). Real estate mutual funds: abnormal performance and fund characteristics. *Journal of Real Estate Portfolio Management*, 6(3), 239–247.
- Elton, E. J., Gruber, M. J., & Blake, C. R. (1996). Survivor bias and mutual fund performance. *The review of financial studies*, 9(4), 1097–1120.
- Evans, R. B. (2010). Mutual fund incubation. *The Journal of Finance*, 65(4), 1581–1611.
- Rodríguez, J. (2007). A Portfolio's Country Exposure Management: The Case of Latin American Mutual Funds. *Emerging Markets Finance and Trade*, 43(2), 5–18.

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